M.P Prasad

Sr Gen AI Developer (RAG's – GenAI – AgenticAI)

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Summary: Experienced Data Scientist with 5.6+ years **(3.5+ in Gen AI)** in developing and implementing AI-agentic, RAG, LangChain, LangGraph, and LLM-driven applications for NLP tasks. Skilled in integrating vector storage, SQL databases, and cloud platforms like AWS and **Azure**, I built intelligent data-driven solutions also designing, optimizing, and deploying scalable AI workflows, with expertise in end-to-end project delivery that drives automation, efficiency, and impactful business outcomes.

Professional Experience Details:

- NTT Data Info Pvt Ltd || Bangalore || Feb 2025 to (Present) as a **Sr Gen AI developer** (Under NLB Services Pvt Ltd)
- ExamRoom.AI India Pvt Ltd || Bangalore || Sep 2023 to Dec 2024 as a Sr Data Scientist
- NV Data Systems Pvt Ltd || Bangalore || June 2020 to Aug 2023 as a Data Scientist

Technical Stacks:

Programming: Python • NumPy • Pandas • Scikit-learn • NLP • SQL • TensorFlow • Keras • PyTorch • OpenCV • CUDA
Machine Learning: Linear Regression • Logistic Regression • Decision Trees • Random Forest • Naive Bayes • K-NN • K- Means • Clustering Algorithms • Boosting Algorithms • Principle Component Analysis • Ensemble techniques.
Deep Learning: Deep-NN • RNN • CNN • LSTM • Transformers • Sparks-Hadoop • GANs • YOLO • MongoDB •
LLM's & Generative AI: Large Language Models(LLM's) • LLM's-Fine Tune-Testing-Merging • Pre-training • GGUF • PEFT's • LoRA • Q-LoRA • Prompt-Eng • DSPy • Text Generation • Unsloth-Training • Machine Translation • AWS • SQL
Management Tools: Project management (Jira) • GitLab • Microsoft Teams • Docker • Kubernetes • CI-CD • Azure Devops
Generative AI Tools:- AI-Agents, Lengchain, LangGrah, Azure OpenAI & Rag, ChatGpt, ADK, MCP, GCP, Vertex-AI

Education Details:

- Bachelor of Technology in Electrical & Electronics Engineering @ Jawaharlal Nehru Technological University-2012

Certification Courses:

- IBM-Rag-and-Agentic-Ai Professional certificate from Coursera with IBM.
- Large Language Model Operations (LLMOps) Specialization from Coursera with Duke University.

Responsibilities:

- Pre-processed structured/unstructured data from SQL and other sources for LLM, RAG, and agentic AI training.
- Implemented RAG workflows with vector storage solutions Chroma, enabling efficient semantic search and knowledge retrieval. Enhanced contextIal text generation with Azure openAI models to improving accuracy and latency.
- Built and fine-tuned Large Language Models (Mistral, Llama) for NLP tasks like summarization, Question-Answering systems.
- Developed Agentic-AI workflows leveraging LangGraph for orchestrating multi-step reasoning, and intelligent task execution.
- Optimized models into GGUF formats (Q-LoRA, PEFT) for efficient inference, lightweight deployment, and faster execution
- Integrated LLM pipelines with SQL-driven automation applications and enterprise data sources, enabling intelligent query handling, automated reporting, and enhanced decision-making efficiency across business workflows.
- Collaborate with cross-functional teams to deliver scalable AI solutions & alignment with business goals and measurable impact.

PROJECTS:-

Project: Text Gen for Various (4) Types of Question Gen Using LLMs (Mistral-7B) (Nov-23 to Sept-24)

LLM's used:

- Python
- Mistral-7B-instruct
- Finetune & Pretrain
- Langchain, GGUF's

- Developed a sophisticated text generation system to create five types of questions using the Mistral-7B Instruct model & also I used different LLM's for various question model.
- Prepared and collected a sufficient large dataset and performed pre-processing also prepared finetuned Dataset format to ensure high relevance and contextual accuracy.
- Fine-tuned model with the pre-processed data, optimize it for question generation tasks.
- Integrated the fine-tuned model into an application, enabling dynamic question generation based on user input.
- Conducted extensive testing and validation to ensure the model's performance and reliability in various scenarios.

Project: Object Detection & Tracking with YOLO (oct-24 to dec-24)

Model's used:

- Python / C++
- YOLO/ONNX
- Retina-Face
- Resnet / Face-net

- Performed extensive data collection and preprocessing to train the model, ensuring accurate and performed Object detection for 34 suspected classes
- Deployed model on cloud platforms, enabling efficient and scalable translation services.
- Optimized YOLO model using and hyperparameter, achieving improve accuracy & faster inference.
- Implemented object tracking with OpenCV, enhancing detection performance for video streams.

Project: Speech Emotion Recognition (Nov-22 to April-23)

Model used:

- Python
- Wave2vec model
- Finetuned
- Deep learning

- Developed a speech emotion recognition system utilizing the Wav2Vec model to leverage state-of-the-art speech processing techniques. Collected and processed audio data from four diverse datasets to train the model, ensuring various emotional expressions and contexts.
- Fine-tuned the Wav2Vec model on the pre-processed data, achieving high accuracy in detecting and classifying emotions from speech.
- Integrated the trained Wav2Vec model into a customer service application, enhancing real-time emotion recognition and response. This improved user experience & increased customer satisfaction.

Project: Document Processing & RAG Pipeline using Azure Document Intelligence (Feb-25 to June-25)

Tools used:

- Python, Pyodbc
- Azure-OpenAI
- Flask, Bs4, fitz
- SQL Database
- ChromaDB, Pypdf

- Built an intelligent document processing pipeline to extract key-value pairs and text from diverse formats using **Azure Document Intelligence** (Form Recognizer)
- Stored outputs in **SQL** & **vector databases** for semantic search, enabling downstream AI workflows.
- Integrated processed data into a **RAG application** for context-aware query answering
- Developed **Flask APIs** and optimized the pipeline for scalability, security, and high-volume document handling.
- Automated end-to-end document ingestion and retrieval workflow, improving processing efficiency and reducing manual effort in enterprise use cases.

Project: Document Extraction Agent & Automating with multi-Agentic system (June-25 to Sep-25)

Tools used:

- LangGraph
- Azure, Cromadb
- Flask, SQL

- Developed a **document extraction agent system** using LangChain and LangGraph, enabling to processing of input files, SOPs, and prompts to extract structured and unstructured data.
- Stored and managed extracted data in **SQL and Chroma vector databases**, integrating with a **RAG-based retrieval system** for context-aware query answering and intelligent information access.
- Finally, this is integrated with a multi-agent AI system for dynamic use across various applications.

Project: Heart Arrhythmia detection by using MIT-BIH ECG Bio signal (Sept-21 to Aprl-22)

Algorithms used:

- Python
- LSTM-CNN
- MIT-BIH data
- Deep learning

- Developed an arrhythmia detection system utilizing the MIT-BIH ECG bio signal dataset to identify and prevent heart-related and cardiovascular diseases.
- Implemented advanced signal processing and machine learning techniques to accurately detect various arrhythmias, enhancing early diagnosis and intervention.
- Conducted comprehensive validation and testing to ensure the system's reliability and accuracy, contributing to improved patient outcomes and preventive healthcare measures.

➢ **Additionally, I have worked on several other projects during my tenure that are not listed here for brevity.**

Declaration:

I hereby declare that all the information given above is true and correct to the best of my knowledge.

Date: Thanks & Regards,

Place: (M Penchala Prasad Rao)